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Research Paper

Optimal tuning of PID controllers focused on energy efficiency in a multivariable and coupled system. Case study: An AHU in the biopharmaceutical industry☆

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ABSTRACT

The main contribution of this work is a methodology for the optimal multi-criteria tuning of commercial PID controllers (more complex and with more parameters than those used in previous research) applied to the control of multivariable processes with interacting control loops. The main goal is to obtain a tuning that achieves, with the lowest possible energy consumption, the desired behavior of the controlled variables. The methodology, based mainly on the adequate modelling of the control system, the choice of the optimization algorithm and the construction of an appropriate cost function, is applied to the tuning of controllers for an air handling unit (AHU) that air-conditions clean rooms in a biopharmaceutical industry. This research also provides a comparison between different cost functions based on performance indices used in the literature. The performance indices are adapted to the case of controlling one variable with two setpoints and a cost function is proposed that achieves better results in the case study addressed than other cost functions used in previous works. The methodology is validated in a real simulation scenario, in which the behavior of the control system when applying the calculated optimal tuning is compared with the behavior of the control system with the current tuning, showing that optimal tuning reduces the energy consumption by 67 kW/h, consuming 23.35 % less energy than the current tuning. Additionally, the ITSE performance indices were reduced by 99.99 % and 76.25 % for relative humidity and supply air temperature, respectively, indicating much better setpoint tracking.

1. Introduction

Given the high percentage of energy consumption in buildings that corresponds to heating, ventilation and air conditioning (HVAC) systems, the optimization of these systems, focused on their energy efficiency, has been of great interest to researchers and engineers. This optimization has been approached both from the point of view of system design and system operation. An example of the first approach is the work of Delač et al [1], which addresses the optimization of a building and its HVAC system from the optimal design, in the context of building refurbishment. The HVAC system simulations were performed in Trnsys software, while the non-dominated sorting genetic algorithm, implemented in Python language by using JEPlus + EA interface, was the optimization method employed to minimize the overall cost and primary energy consumption by proper sizing of HVAC system components, selection of the most efficient HVAC system technology and the most suitable parameters of energy efficiency measures.

In the second approach, the work aimed at making control algorithms effective in ensuring that requirements such as thermal comfort and air quality are met, while at the same time ensuring that the system operates in a way that consumes as little energy as possible, is remarkable. For example, Fang et al. [2] designed a Deep Q-learning (DQN) based multi-objective optimal control strategy for the generation of optimal chilled water supply temperature and chilled water flow temperature setpoints, to minimize energy consumption while maintaining

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Nomenclature

Abbreviations

AHU	Air Handling Unit
CF	Cost Function
HVAC	Heating, Ventilating and Air Conditioning
IAE	Integral of Absolute Error
ISE	Integral of Squared Error
ITAE	Integral of Time Absolute Error
ITSE	Integral of Time Squared Error
PID	Proportional-Integral-Derivative
PLC	Programmable Logic Controller
SCADA	Supervisory Control and Data Acquisition
TV	Total variation

Variables

TSu	Supply temperature
HuSu	Supply relative humidity
TEx	Return temperature
HuEx	Return relative humidity
SpTSuHi	High level setpoint for supply temperature
SpTSuLo	Low level setpoint for supply temperature
SpHuSuHi	Setpoint for relative humidity
U_VE	Control signal to heating coil
U_V3V	Control signal to cooling coil
K_p	Proportional gain
T_i	Integral time
T_d	Derivative time
T_f	Time constant of first order filter of derivative action
T_t	Time constant for back calculation
N_z	Neutral zone
T_r	Rise time
I_{max}	Saturation value for integral part
E_{heat}	Energy needed for heating
E_{cool}	Energy needed for cooling
P_{heat}	Power consumed by the bank of electrical resistors
P_{cool}	Power consumed by the cooling coil
ITSEh	ITSE value for relative humidity
ITSEt	ITSE value for temperature
ITSEh_n	ITSE scaled value for relative humidity

adequate indoor air temperature. This study developed an EnergyPlus-Python co-simulation testbed for performance evaluation of the DRL (deep reinforcement learning) control strategy in a multi-zone building VAV system. To the development of optimal control is also dedicated the research of Lee et al. [3]. The control object was the economizer of an AHU and, using a neural network model, evaluated by Trnsys-Matlab co-simulation, they were able to meet the thermal comfort and air quality requirements with minimum energy use. The optimal control proposed in this work resulted in energy savings of 20 %. A broader view of the optimization problem presented by HVAC systems is presented by Wang and Zhao [4], who show it as a hybrid problem, whose solution is approached by simultaneous consideration of the temporal and spatial characteristics. Then, they propose an adaptive hybrid differential evolution algorithm to solve the optimization problem based on the spatio-temporal characteristic (TSO), simultaneously determining the optimal timing (ON-OFF sequence) and setpoints.

It is very common to find in literature that control algorithms that implement optimization methods aimed at energy efficiency focus on the generation of optimal setpoints. A control algorithm whose application in the optimization of the operation of HVAC systems has been increasing for some years is the model-based predictive control (MPC). An example of the use of predictive control applied to the generation of optimal setpoints is the one developed by Heidary and Braslavsky [5], who propose this type of controller for the control of the cooling system of a commercial building. This work optimized the operation and planning of the cooling system (planning and operating capacity of chillers, operation of cooling towers, among other variables), achieving savings of 25 % compared to the operation of the existing control system. The control algorithm required the solution of a nonlinear integer-mixed programming problem, transformed by linearization, discretization, among other techniques, into a mixed integer linear programming problem, implemented in the Julia programming language and solved using the Gurobi solver. [5,6].

However, this type of controller has also been used to guarantee optimal setpoint tracking, as proposed by Wang et al. [7], to minimize the total energy consumption and cost of a complete chiller-AHU system. In this work the MPC leads to the minimum power consumption of the fan, pump and chiller, while maintaining the same room temperature. Unlike other works where the predictive controller is used to generate optimal setpoints, this study uses the MPC in substitution of PI controllers. However, this practice of replacing PID controllers with predictive controllers is less common.

PID controllers are at the low levels of all complex control systems, due to their reliability and robustness, which allows them to perform well even in the control of processes with non-linear and variable dynamics. However, studies shows that a large part of them operate in manual mode, while most of those operating in automatic mode perform poorly, due to poor choice of their parameters [8]. Despite this, and the existence of newer and more advanced control techniques, they remain the most popular controllers in industry, used in around 90 % of industrial processes, stimulated by the simplicity of their algorithm and ease of re-tuning, in addition to the above-mentioned strengths. However, the success of PIDs depends on the accuracy of the system model and its variables, and their tuning continues to be an important area of research, with a growing interest shown by the scientific community in the tuning of their parameters [9].

1.1. Optimization methods for tuning PID controllers

PID controller tuning strategies have been classified in different ways by different authors. Those strategies based on optimization methods are characterized by the controllers' parameters being set from the off-line solution of an optimization problem, through the application of different methods, such as genetic algorithms (GA), particle swarm optimization (PSO), differential evolution (DE), among others [9].

Many studies focus on tuning the parameters of a single controller, as is the case of Almabrok et al. [10], who used the Big Bang-Big Crunch (BB-BC) evolutionary algorithm, which was faster than the PSO and GA algorithms for tuning the parameters, in this case gains, K_p, K_i and K_d; Kouassi et al. [11], who used the Ant Colony Optimisation (ACO) algorithm to tune the aforementioned parameters of a PID controller; and Fiducioso et al. [12], who tuned the Kp and Ki parameters of a PI controller using the Safe Contextual Bayesian Optimization algorithm.

Meanwhile, the tuning of more than one controller has been the subject of research works such as those conducted by Oonsivilai and Marungsri [13], who, using the Adaptive Tabu Search algorithm, tuned the three gains of two PID controllers, work that was reduced only to the tuning of three parameters, since both controllers were considered with identical parameters; A.P. Wemhoff [14], where the gradient descent method was used to calibrate the three gains of three PID controllers simultaneously, (nine parameters to be tuned in total); Copot et al. [15], who proposed multi-objective particle swarm optimisation (MOPSO) to tune the gains of 8 PIDs, but independently, whereby each optimisation process found the optimal values of the three gains of each PID controller; similarly, the work of Ambroziak and Chojecki [16], dedicated to the tuning of 4 PID controllers, making use of the Fuzzy Self-Tuning PSO (FST-PSO) algorithm for the selection of their parameters, tuned the gains of each controller individually, as the controllers were integrated in a control scheme in which only one operates at a time.

It is noticeable in the study of the bibliography performed in this research that most of the works dealing with the optimal tuning of controllers used the classical textbook algorithm, or very basic algorithms of more advanced controllers. These algorithms do not consider all the parameters that can be tuned in commercial controllers, which are more complex, with nonlinearities, and can have more than twice the number of parameters than most PID control algorithms used in research dedicated to optimal tuning of PID controllers. On the other hand, most of the works tuned a single controller or several, but without considering the interactions between control loops.

Multivariable control systems with coupled control loops are a challenge when tuning PID controllers, since the control signal from one loop can disturb another control loop, or, in the worst case, all the control loops can disturb each other. The difficulty of this task is increased when the tuning of the controllers must ensure that the control system meets more than one requirement, which can lead to the solution of a multi-criteria optimization problem.

The aim of this work is the multi-criteria optimal tuning of commercial PID control algorithms (of higher complexity than those addressed in the literature), which are part of a multivariable control system with coupled control loops. This research was motivated by the need to find the optimal tuning of the PID controllers that regulate the temperature and relative humidity of the air supplied by an air handling unit (AHU) to a set of clean rooms. The controllers to be tuned have seven parameters each. Given the coupling of control loops, it is proposed the simultaneous tuning of the three controllers, which implies finding a set of twenty one parameters that minimize the deviations of the controlled variables from their respective setpoints (generated by another control system which is not the focus of this work), to do this with the minimum energy consumption and with control signals as stable as possible.

The contributions of this work are: 1) a general method for simulation based optimal tuning of commercial PID controllers, 2) a procedure for multicriteria tuning of PID controllers suitable for simultaneous tuning of multiple controllers involving a large number of parameters, compared to the number of parameters that are simultaneously tuned in previous research, some of which are cited here, and 3) a cost function based on the integral of the time multiplied by the quadratic error, the total variation of the control signals, the energy consumption models of the heating coil and the cooling coil, the scaling of each term and adjustable weights for them, according to the proposed methodology. This research also adapts the ITSE computation to the case of controlling a variable with two controllers following different setpoints. The proposed cost function is evaluated and compared with nine other cost function variants.

It is shown through different simulation scenarios that, by using the proposed cost function, better optimization results are obtained for the case study addressed in comparison with other cost functions used in previous research works. Those scenarios also validate the proposed methodology, showing a much better performance of the control system with the PID tuning obtained, especially when comparing with the current tuning in a real scenario where setpoints were followed with more accuracy while reducing energy consumption in more than 23 %.

The main limitations of this work are: 1) the results were obtained by simulation, validation in the real system is pending to prove the effectiveness of the proposals of this work; 2) the comparison could be richer, incorporating other performance indexes and other optimization methods and 3) the success of the methodology is dependent on the quality of the control system model and it is not always easy to obtain all the models required.

2. Metodology

2.1. Proposed general method to perform simulation-based tuning of commercial PID controllers

A variety of PID control algorithms have been developed throughout history. A collection of the main algorithms can be studied in the work of Astrom, K. J. and T. Häglund, which describes several modifications to the classical PID algorithm, such as the inclusion of derivative action filtering and the inclusion of anti-windup methods [17]. It is crucial to know the control algorithm being tuned if this tuning is performed by simulation. Tuning by simulation a PID controller based on an algorithm (model) that does not match the one implemented in the real controller may imply that, when applying the parameters of the PID model used in simulation to the real controller, the control system does not respond in exactly the same way as it does in simulation or even that its response is completely different or becomes unstable [18].

The use of classical methods, such as Ziegler and Nichols, for tuning PID controllers in industrial processes is limited by the complexity that PID algorithms in commercial controllers can reach. These algorithms, in addition to nonlinear dynamics, can include a number of tuning parameters not conceived in classical methods, which, on the other hand, do not guarantee to find the optimal PID controller parameters [19] and a fine tuning procedure usually follows the application of tuning formulas. That is why, in the last decades, a considerable amount of research related to PID controllers has been focused on the optimal tuning of their parameters.

The off-line tuning of PID controllers using optimization methods, which are based on the simulation of the control system to obtain the optimal tuning parameters according to a given cost function, presents different challenges. First, the development of a model of the control system, which is usually composed of several submodels depending on the elements that constitute the control system. These submodels can be grouped into: 1) model of the process to be controlled (including actuator and sensor models) and 2) model of the controller (which may contain one or more control algorithms, depending on the complexity of the control system).

If system identification is used as a method to obtain the model of the process to be controlled, then it is obtained a model that includes not only the dynamics of the process, but also of sensors and actuators. In the case of the controller, several scenarios can be presented at the time of modeling for subsequent tuning by simulation and application of this tuning to the actual controller:

1) Who tunes the controller was who designed and/or programmed the control algorithm: this is the case of programming a PID algorithm taken from the literature in a microcontroller or other device, where the programmed algorithm is known and can be reproduced in simulation software.
2) Who tunes the controller did not design or program the control algorithm, but knows, if not the programmed algorithm, at least a mathematical model to which this algorithm responds: this is the case of using commercial controllers, such as PLCs, when the technical documentation provides sufficient information of the control algorithm (such as a model in the time domain or Laplace domain) for simulation.
3) Who tunes the controller does not know the programmed control algorithm: in this case, or technical documentation is not available or does not provide sufficient information of the control algorithm.

Work on simulation-based PID controller tuning usually assume that the control algorithm or the dynamic model that describes the controller is completely known and directly go to apply a tuning method. However, in practice, when faced with the task of tuning a commercial controller,

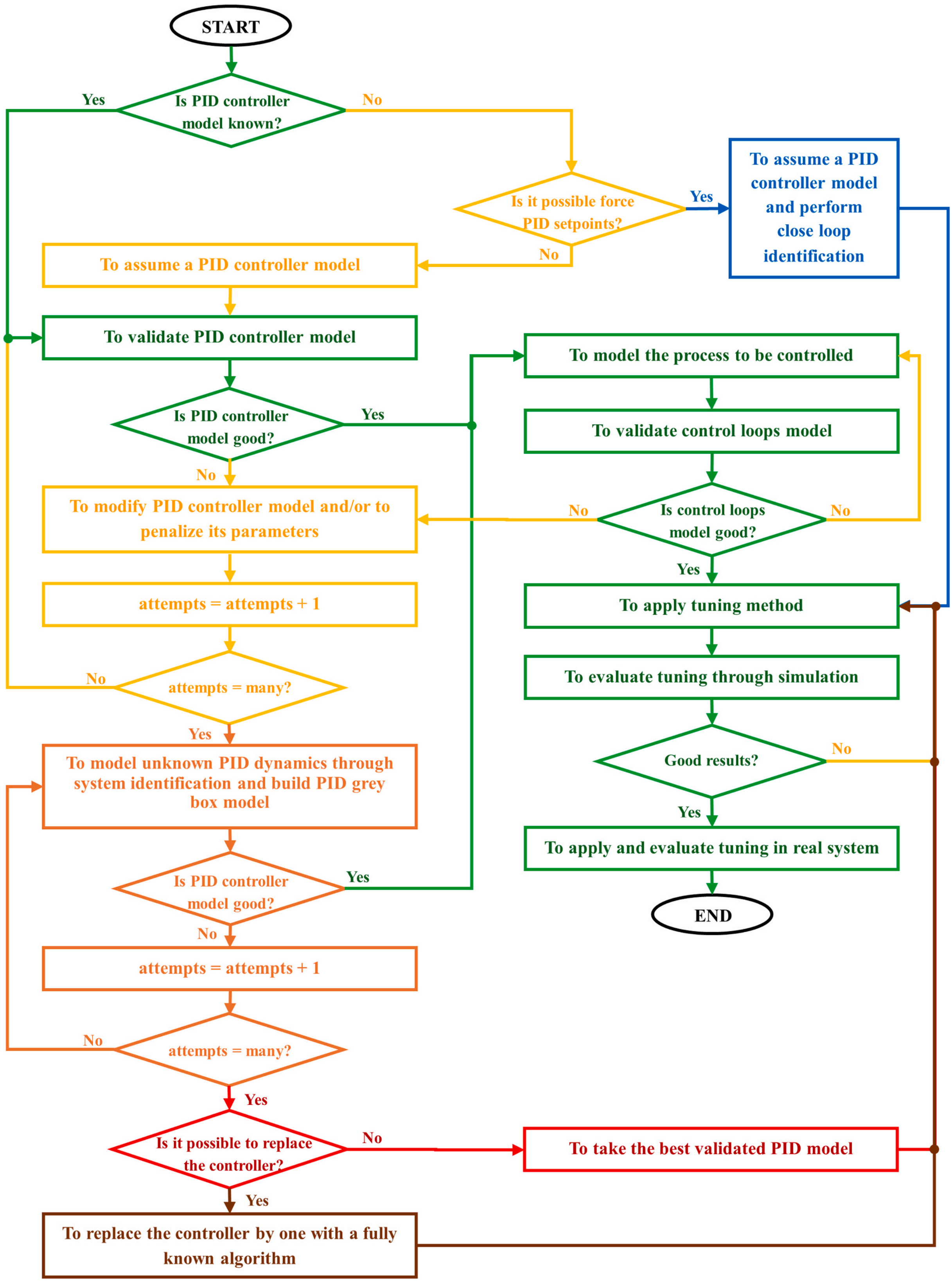

Fig. 1. Proposed general method to perform simulation-based tuning of commercial PID controllers.

as explained above, the control algorithm is not always known. This paper proposes a general method (illustrated as a flowchart in Fig. 1) to perform simulation-based tuning of commercial PID controllers to ensure that a valid and reliable model of the control system whose controllers will be tuned is available before using any computational tuning method, so that a response of the real system can be expected, when applying the obtained tuning, very similar to that obtained in simulation.

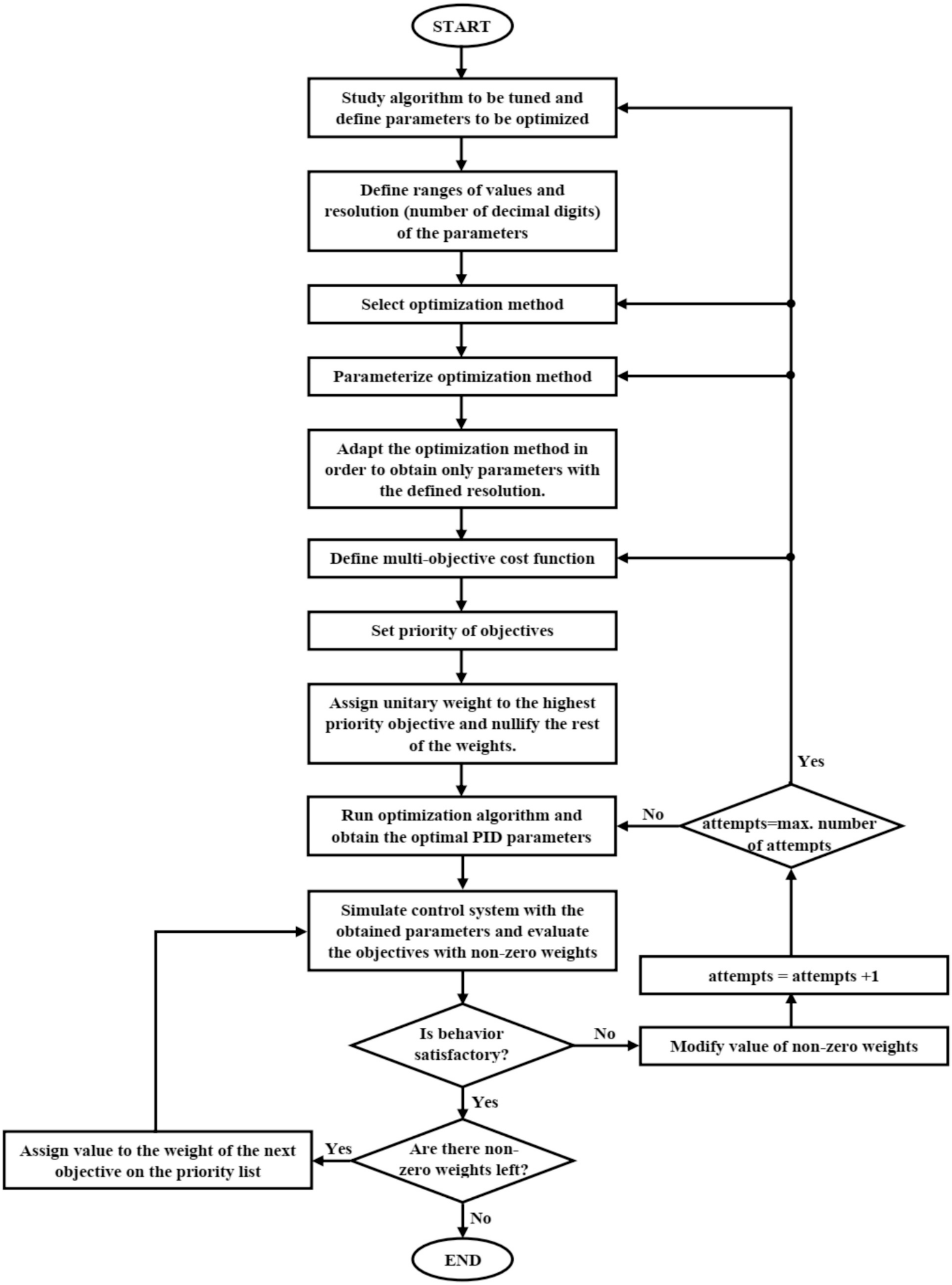

Fig. 2. Proposed procedure to apply multi-criteria tuning of commercial PID algorithms.

The general method described in Fig. 1 highlights in color different scenarios that can occur. The best case is to know exactly the algorithm programmed in the controller or a model given by the manufacturer. Following the sequence of steps marked in green, it is always advisable to validate the controller model anyway, since a model is not necessarily an exact representation of the real system. The step consisting of modeling the process to be controlled includes the validation of that model. Then, as the tuning is based on simulating the closed control loop, in order to obtain the PID parameters that offer the best behavior, the closed control loop must be validated (connecting the PID controller and process models and closing the feedback loop), since, even if an accurate model of the controller is available, in the case of the process there is still a model that is an approximation with a certain quality.

If neither the algorithm nor a model of the PID controller is known, its mathematical model must be obtained, for which it must be evaluated if it is possible to access the setpoint and force their values (depending on the design of the control system and the level of access of those who operate it, this may be impossible for internal controllers of cascade schemes). If it is possible to modify the setpoint values, a closed-loop identification can be applied and the control loop model can be obtained, a technique for which several methods have been proposed, as evidenced in Ref. [20–22].

If it is not possible to modify the setpoint values but to record them, it is then possible to try to find a good enough model of the controller. To do this, firstly a model from literature can be assumed. As there are several, it is important to gather as much information as possible about the control system by studying its technical documentation, including documentation of similar products from the manufacturer.

To validate the assumed model, it is suggested to sample data from the setpoint, the PID control signal and the controlled variable. Then, by simulation, the PID controller model can be fed with the data collected from the setpoint and the controlled variable to compare the similarity of the simulated control signal with the real one. If the result of this validation is not good, it can be tried penalizing the parameters of the PID model assumed, modify the model or replace it with another and validate again. If this is not enough to obtain a good result, the commercial PID may be much more complex than the assumed models. In this case, an alternative is to model the unknown dynamics of the commercial controller by building a gray model, as suggested in Ref. [23].

If it is not possible to develop a model with sufficient quality of the closed loop control, it should be considered the possibility of designing the control system in a different hardware whose algorithm could be known (another commercial controller or a different hardware). Otherwise, if the validation of the closed loop is not good enough but is not "terrible", it can still be evaluated the possibility of applying the optimization method to tune the parameters in simulation, based on the best model obtained, test them in the real PID and evaluate the results (if this procedure does not represent a risk for the industrial process).

Whatever the path in Fig. 1 to which leads the system whose controllers are required to tune, all converge in the application of a tuning method. In control engineering problems it is usually expected that the response of the controlled system meets several requirements (such as a fast response without oscillations, a small overshoot, an economic control, among others) often in conflict. For this reason, one approach to find a PID controller tuning with a desired trade-off is to define an optimization statement to solve a multi-objective problem [24].

3. Procedure to apply multi-criteria PID tuning of commercial PID algorithms

Optimization is the most efficient approach to tuning PID controllers for nonlinear or high-order processes and allows the problem addressed here to be approached in two fundamental ways: 1) aggregate the objectives into a single objective, introducing weight factors to give different priority to different objectives, or 2) solve a multi-objective optimization problem with a Pareto-based method. The first variant has the advantage of solving a simpler problem, at the cost of being necessary iterations in the design to obtain an acceptable trade-off [25]. The second variant results in a Pareto front approach, a curve composed of conflicting solutions to the problem objectives, where each solution is considered valid, since there is no best solution across all objectives [26], unlike single-objective optimization, where the application of search algorithms results in a single solution [27].

Although the method of aggregating different objectives into a single objective does not allow finding all the points of the Pareto set [25], it is frequently used in research related to the optimal tuning of PID controllers. The fact that the weights are selected by trial and error iteratively until an acceptable trade-off between the different objectives is achieved is its main drawback. Since the different objectives included in the cost function may have different ranges of values, units of measurement and standard deviations, their contribution to the cost is not uniform. A solution to this is the calculation of the weights proposed in Ref. [28], based on the calculation of the percentage contribution of each term, for which the average of the Pareto solutions is used, statistically ensuring an equivalent contribution of the terms of the multiobjective function.

On the other hand, in the case of the multiobjective approach, more complex versions of methods that traditionally solve single-objective problems are required to solve multiobjective problems. While it is negative to have many objectives when applying the objective aggregation approach, since more weights must be tuned, it is also a drawback in the Pareto-based multiobjective approach. Many-objective optimization instances, where more than five objectives must be optimized [24], represent a particular challenge for multi-objective optimization algorithms, as convergence and propagation capability often conflict [26]. Then the objective aggregation approach was chosen as a method to achieve the main goal of the presented study case, since, first of all, it is desired to evaluate and select the objectives to be taken into account in the cost function to provide the best results and for that purpose this approach is sufficient. Thus, the procedure described in Fig. 2 is proposed to be applied for the optimal tuning of PID controllers in this case.

For the tuning procedure in Fig. 2, it is very important to study the control system and the controller documentation to know which parameters should be tuned, the range of values and the resolution of each one. The latter is an important detail, since the optimization method applied should not result in parameters with a resolution impossible to apply in the real controller. For example, if the time values can be specified down to the order of milliseconds, optimal tuning should not be obtained with times specified down to microseconds, since to apply them in practice it would be necessary to truncate the parameter values and in that case the result might not be the same as the one obtained in the optimization process.

The selection of the ideal optimization method is not the subject of discussion in this paper. In principle, this can be selected based on previous experience with similar problems or based on methods recommended by literature. In the summary and classification of PID controller tuning methods presented in Ref. [9] it can be seen that the computational optimization techniques usually used in these cases are based on metaheuristics, and a great diversity of methods have been evaluated, which can be mainly grouped into: swarm intelligence algorithms, evolutionary algorithms, stochastic algorithms, neural algorithms, immune algorithms, physics related algorithms, probabilistic algorithms and neuro-fuzzy algorithms.

The construction of the cost function depends on the aim of the PID controller tuning. To meet specifications of the time response of the system several authors have based the cost function on performance indices calculated from the error between the controlled variable and the setpoint. The most frequently used, evidenced in Ref. [10,11,29], are IAE (integral absolute error), ITAE (integral time absolute error), ISE (integral square error) and ITSE (integral time square error), represented in Eq. (1) to (4) respectively, in their continuous and discrete variants.

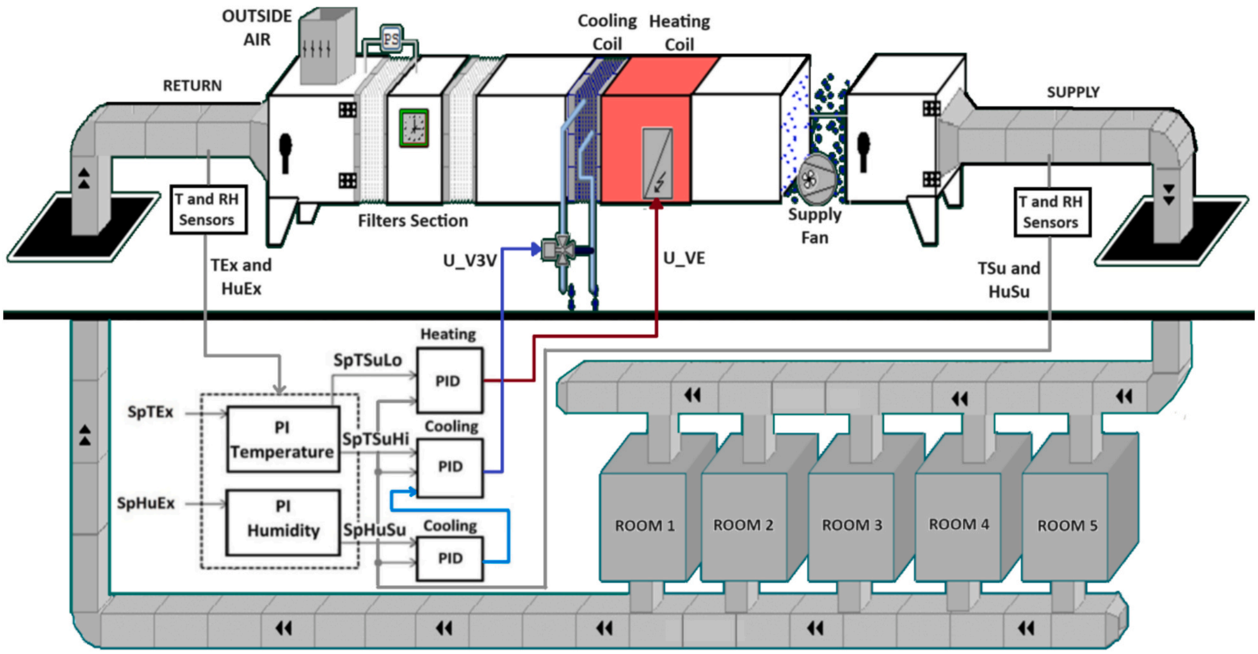

Fig. 3. AHU, rooms and studied control loops.

$$ISE = \int_0^\tau e^2(t)dt \approx \sum_{k=0}^{M} \frac{e^2(k) + e^2(k-1)}{2} T_s \tag{1}$$

$$ITSE = \int_0^\tau te(t)^2 dt \approx \sum_{k=0}^{M} k \frac{e^2(k) + e^2(k-1)}{2} T_s \tag{2}$$

$$IAE = \int_0^\tau |e(t)|dt \approx \sum_{k=0}^{M} \left| \frac{e(k) + e(k-1)}{2} T_s \right| \tag{3}$$

$$ITAE = \int_0^\tau t|e(t)|dt \approx \sum_{k=0}^{M} k \left| \frac{e(k) + e(k-1)}{2} T_s \right| \tag{4}$$

Although minimizing a function of the error between the controlled variable and the setpoint is usually a way to achieve desired characteristics in the time response of the system, some authors have treated separately, as different objectives, each specification of the time response that is of interest, as in Ref. [15,28]. Other authors have considered both performance indices and time response specifications as objectives, as in Ref. [19]. In addition to the specifications of the time response of the control system, the behavior of the control signal has also been considered, including in the cost function indices such as ISCO (integral square of control output), as in Refs. [30,31], and TV (total variation) as a measure of the control effort [32]. These indices are represented in Eq. (5) and (6) respectively.

$$ISCO = \int_0^\tau u^2(t)dt \approx \sum_{k=0}^{M} \frac{u^2(k) + u^2(k-1)}{2} T_s \tag{5}$$

$$TV = \sum_{k=1}^{M} |u_{k+1} - u_k| \tag{6}$$

The minimization of indexes such as ISCO is aimed at reducing the cost of control. On the other hand, TV is a measure of control effort and minimizing it seeks smooth control signal responses, useful when it is desired not to affect the life of actuators with mechanical components that may suffer wear. Also, rapid changes in the control signal are not desirable when the actuator has a large time constant in its response compared to the time in which the control signal changes, so that it would not be able to follow it.

In the case of control cost, when this is manifested in a measurable magnitude, such as energy consumption, this work proposes to use as a performance index directly the calculation of the energy consumed, from the use of an energy consumption model that relates the values of the control signal with the values of energy consumed consequently. In this research, the two variants, the use of ISCO and the use of energy models as part of the cost function, are evaluated to determine which is more suitable for the case study.

As explained above, in this case the cost function is constructed as the weighted sum of the different objectives to be considered. As a method for tuning the weights of the objectives, it is proposed to group the objectives in order of priority. Then, in an iterative process start tuning the weights of the highest priority objectives, nullifying the rest, so as to ensure good performance for the highest priority objectives first. Then, move on to the next objectives in priority, tuning their weights as it is assessed that, while improving performance with respect to these objectives, the possible degradation of the higher priority objectives is within the range of what is permissible. When the best compromise is found, then the weights of the next objectives in priority are tuned.

This procedure is useful when there are many objectives, which may be the case in multiple-input multiple-output (MIMO) control systems, in which one or more objectives are evaluated for each input and output. The disadvantage consists in the trial-and-error application of a tuning, iteratively looking for a good compromise. It may seem that there is not much benefit to be gained if in the end a manual tuning must be done. But, as will be seen in the following sections, this manual tuning, iteratively and orderly, allows optimal tuning of multiple simultaneous controllers, with a total number of parameters more than three times the number of parameters tuned in the cost function.

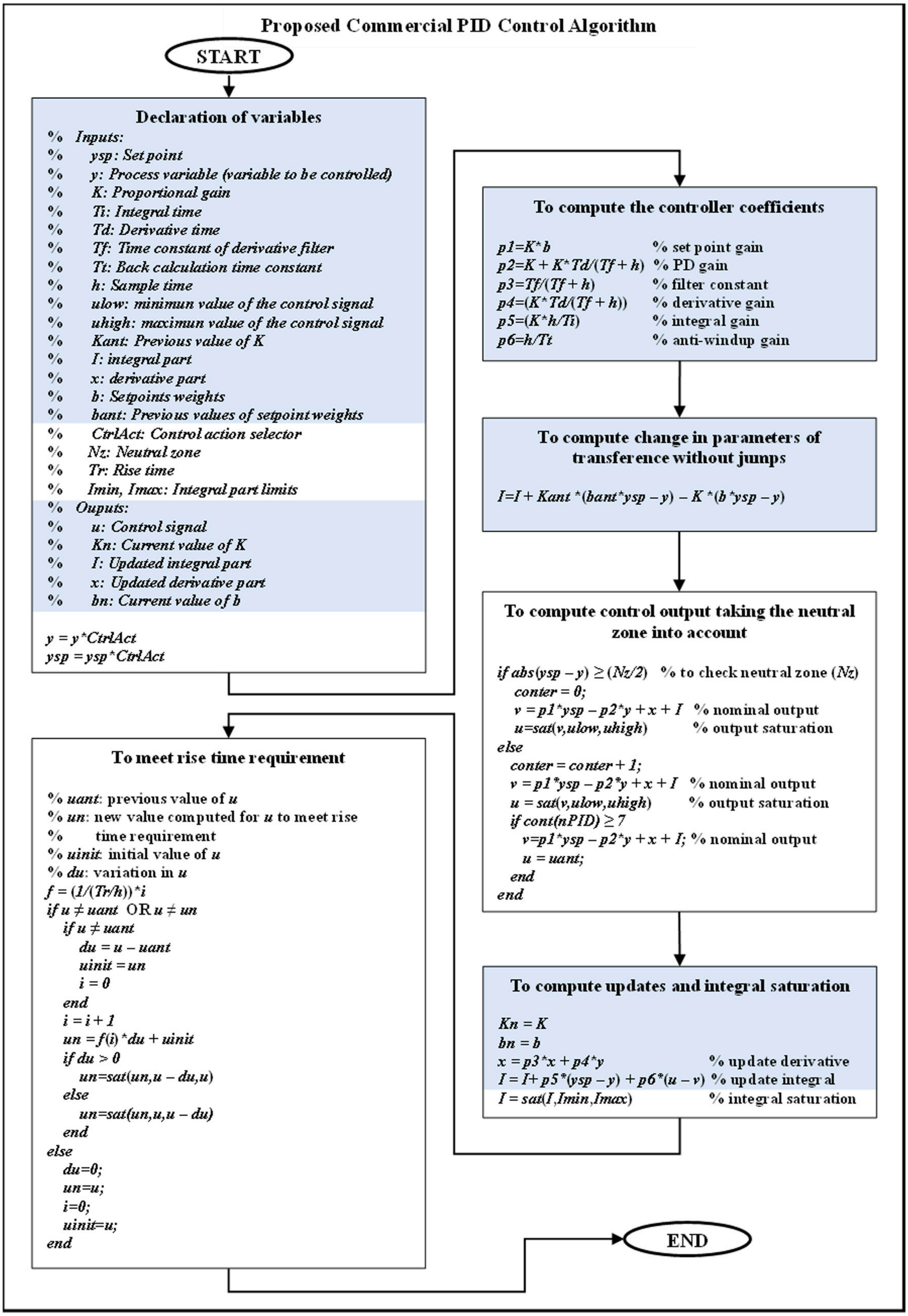

Fig. 4. Proposed commercial PID algorithm.

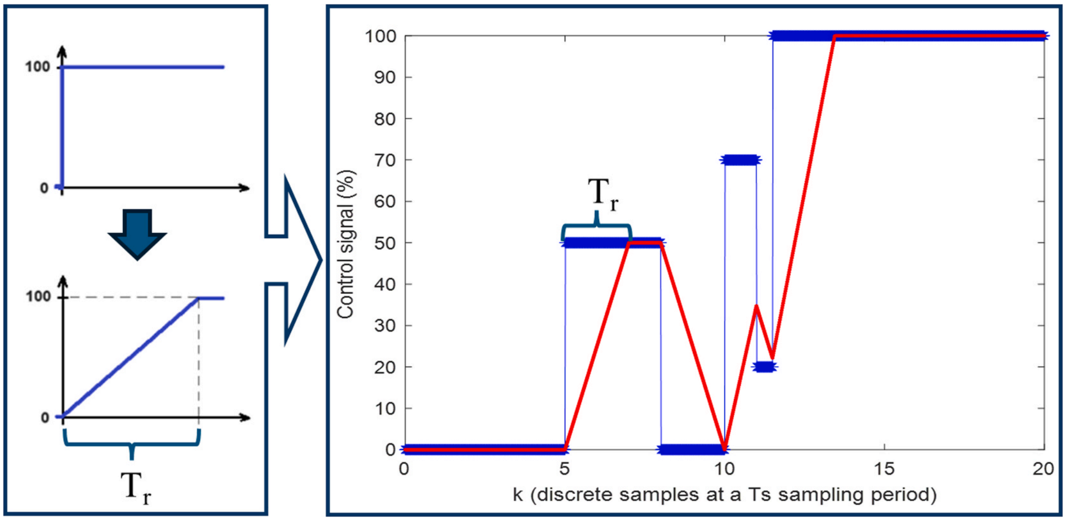

Fig. 5. Effect of the introduction of a rise time (T_r) in the output of a controller.

4. Application of the methodology to the case study: Control of an air handling unit in the biopharmaceutical industry

4.1. Description of the system

The system under study is an air handling unit (AHU), which air-conditions five clean rooms in a biopharmaceutical industry, and its temperature and relative humidity control system. Fig. 3 shows a schematic of the AHU, rooms and the temperature and relative humidity control loops. An internal loop, composed of PID controllers, regulates the supply temperature (TSu) and the supply relative humidity (HuSu). The setpoints for controlling these variables are received from an external control loop, which regulates the return temperature (TEx) and return relative humidity (HuEx). The current control system regulates the temperature by both heating and cooling, while the humidity is regulated only by cooling, setting the minimum value of the cooling temperature control signal.

This work is focused on the optimal tuning of the PID controllers that are part of the internal control loop. These controllers act on the AHU through the cooling and heating coils. The approach of this work is that, if it is desired to optimize the cascade control system, it must first be optimized the internal control loops. Otherwise, it would not be guaranteed that the optimized references, resulting from future optimization work on the external loops, would be fully exploited. Therefore, the aim in this case study is an optimal performance in the tracking of the references by the PID controllers of the internal loops, regardless of whether the references, coming from other controllers, are the most appropriate.

The cooling coil of the AHU consists of a heat exchanger through which cold water at a temperature of 7 °C flows. The water flow is controlled by a three-way valve that receives a signal from the controller (U_V3V) in the range of 0 to 10 V. The heating section is composed of a bank of electrical resistors whose power is regulated by a three-phase power controller, receiving a control signal (U_VE) also in the range 0 to 10 V.

According to the AHU technical documentation, the cooling coil is designed to, by receiving water at 7 °C, bring the inlet temperature from 23 °C to 13.2 °C at the outlet and the inlet relative humidity from 59 % to 96 %. Also, according to the documentation, this coil has a power of 27 kW. The heating coil is designed to raise an inlet temperature from 13.2 °C to 20.5 °C at its outlet and lower a relative humidity from 96 % at its inlet to 60.5 % at its outlet.

Then, the air at the outlet of the heating coil, with a temperature TSu measured in degrees Celsius (°C) and a relative humidity HuSu measured in percentage (%) is blown into the rooms. Sensors in the outlet duct feed these temperature and relative humidity values back to the internal loop PID controllers, which receive the setpoints SpTSuLo, SpTSuHi and SpHuSuHi. The first two are the low-level and the high-level supply temperature setpoints, respectively, represented inside the controller as values in degrees Celsius (°C), the other setpoint is the supply relative humidity, represented inside the controller as values in percentage (in a range 0–100 %). Although the control signals U_VE and U_V3V that are sent to the heating and cooling coils respectively are physical signals in the range 0–10 V, inside the controller these variables are represented in percentages, in the range 0–100 %. This is the magnitude used in the modeling of the control system and subsequently in the control system simulations.

The PI blocks in Fig. 3 that generate the setpoints of the internal control loops are fed back with the variables TEx (return temperature) and HuEx (return relative humidity), measured and represented in the controller by their values in degrees Celsius (°C) and percentage (%) respectively. SpTEx (°C) and SpHuEx (%) are the return temperature and the return relative humidity setpoints respectively. This variables are not of interest for the work developed here.

4.2. Application of the methodology

4.2.1. Construction of a PID model and validation of the control system

In the case study, there is an AHU control system already designed and in operation, which is required to be optimized. As a first step of the proposed methodology, this control system is studied, which has been implemented in a PXC100-ED controller from Siemens, programmed with Xworks software. The SCADA system was designed using the Desigo Insigh product from the same manufacturer. From the technical documentation it was not possible to obtain a model of the implemented control algorithm, only the description of its functionalities, the parameters to be tuned and how some of them affect the system response.

In this case a PID control algorithm was constructed, based on a PID control algorithm obtained from literature (Astrom and Hagglund [17]) with the general characteristics of a commercial PID controller. Then, based on this algorithm, a more complex algorithm was built, adding operations and parameters based on the information available in the technical documentation of the control system. The flowchart in Fig. 4 shows the algorithm resulting from modifying the literature algorithm (shadowed) by adding a neutral zone (N_z), a rise time (T_r) for the control signal, and the option to saturate the integral term (I). The neutral zone (N_z) works in such a way that, once the controlled variable falls within its limits, the algorithm waits seven sampling intervals (according to the

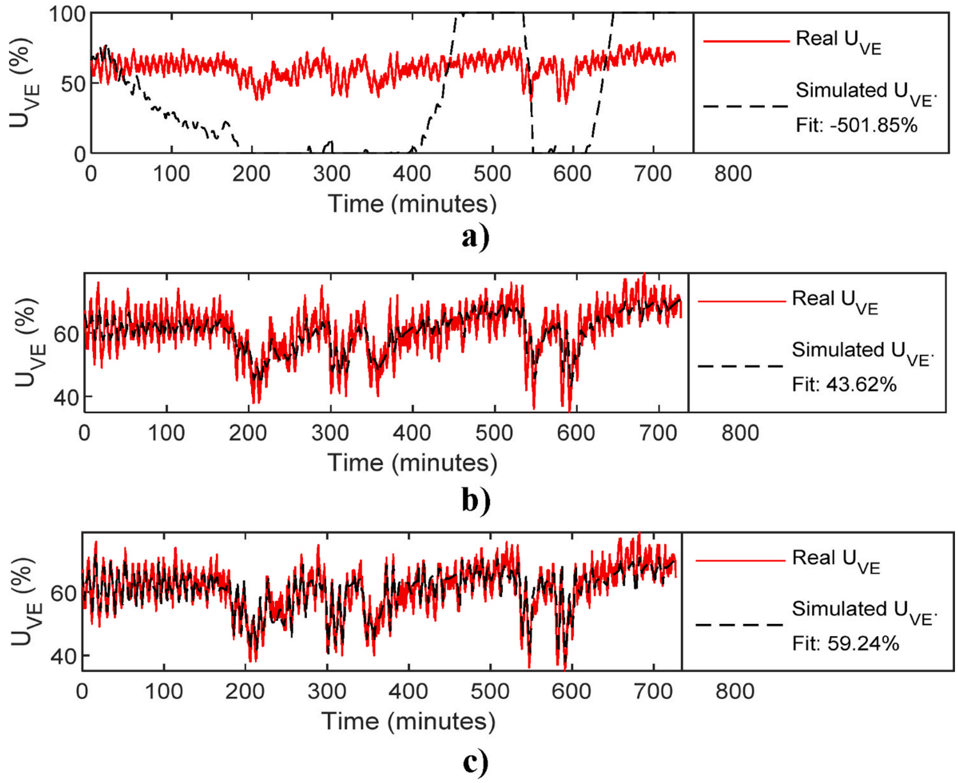

Fig. 6. Results of the open loop validation of the proposed PID algorithm: a) algorithm just as described in Fig. 4; b) algorithm described in Fig. 4 multiplying K_p by 1.6 and T_i by 1440; c) algorithm described in Fig. 4 whose output has been connected to a black box model identified as suggested in Ref. [23].

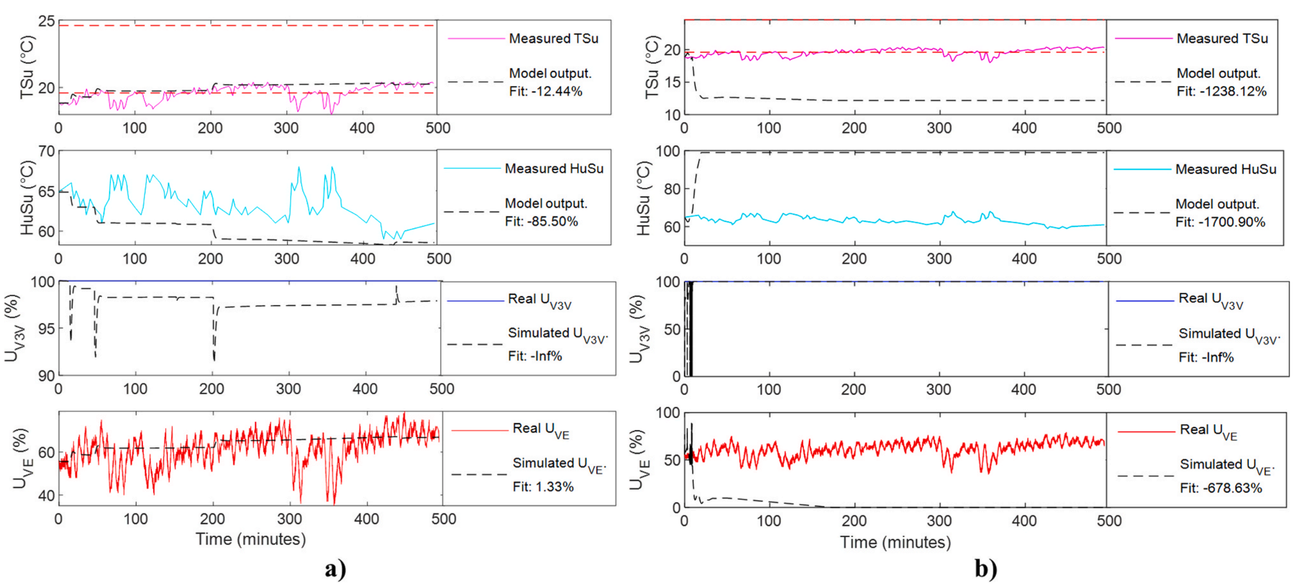

Fig. 7. Results of the closed loop validation of the proposed PID algorithm: a) algorithm described in Fig. 4 multiplying K_p by 1.6 and T_i by 1440; b) algorithm described in Fig. 4 whose output has been connected to a black box model identified as suggested in Ref. [23].

documentation it should be seven cycle times, but since the cycle time of the actual controller is not known, the sampling time was taken instead), after which, if the controlled variable is still in the neutral zone, the control output remains the same. The effect of rise time (T_r) can be clearly seen in Fig. 5.

The results of the validation of the PID model and the control system are shown in Fig. 6 and Fig. 7. First, the open-loop validation of the proposed PID algorithm was conducted, where it was found that this algorithm, when simulated with the same parameters as the PID algorithm implemented in the controller, has a very different behavior (see Fig. 6 a)). However, when penalizing the proportional gain and the integral time, it can be observed in Fig. 6 b) that the similarity between the output of the proposed PID model and the real PID greatly increases. However, the goodness of fit (*Fit*), calculated according to Eq. (7) [33], is still not very good.

$$Fit = \left(1 - \frac{\|y - \widehat{y}\|}{\|y - \overline{y}\|}\right) \bullet 100 \tag{7}$$

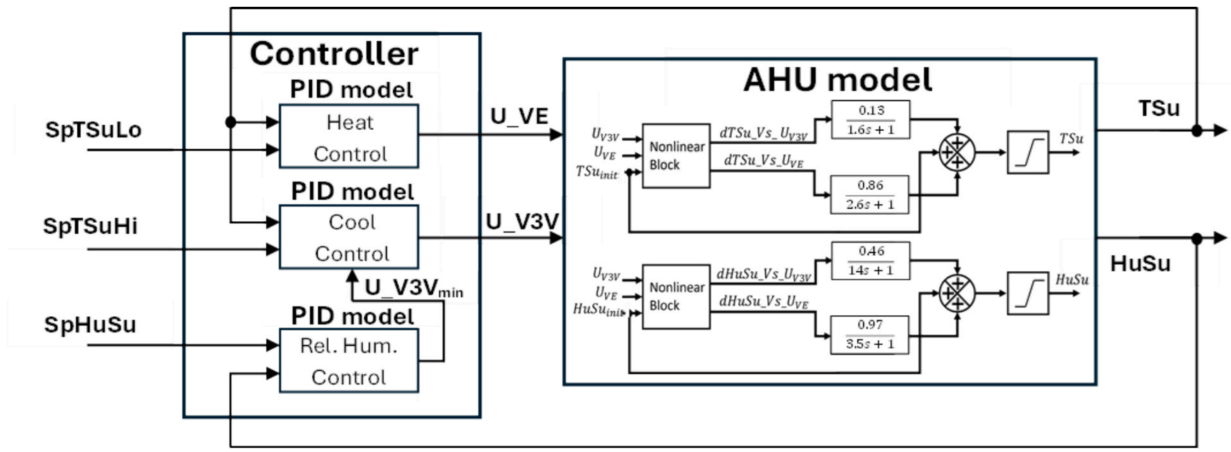

Fig. 8. Block diagram of the control system model (internal loops).

Table 1
PID parameter ranges and resolutions.

Parameters	K_p	T_i (min)	T_d (min)	T_f (min)	N_z	T_r (min)	I_{max}
Range of values	[0.1 1000]	[0 10]	[0 1]	[1.66667•10^{-5} 10]	[0 2]	[0 5]	[50 100]
Resolution	Decimals	Milliseconds	Milliseconds	Milliseconds	Decimals	Milliseconds	Integers

A better *Fit* is obtained by using the gray model proposed in Ref. [23], which is composed of the algorithm in Fig. 4 coupled with a black box model at its output, obtained by system identification. However, in the closed-loop system validation the results are totally different. The gray model, which had performed better in the open-loop validation, has a very poor performance in the closed-loop validation. This, added to the fact that it consumes much more time in the simulation than the algorithm in Fig. 4 with its weighted parameters, makes it unsuitable for use in tuning. The algorithm of Fig. 4, with its weighted parameters, has a behavior that, at least at the beginning, is similar to the behavior of the real system, although the *Fit* value is very poor.

It should be noted that, in the simulation of a real scenario like this one, there is significant uncertainty. For example, the value of the sampling time of the PID algorithm is not really known, it is assumed to be the shortest time between two samples acquired by the SCADA system, which coincidentally is about ten times smaller than the smallest time constant of the AHU model, which is a very reasonable sampling time. Also, the initial values of the controllers' integral terms are not known, which respond to the past of the system, unknown to the simulation. Therefore, given the results in the validation of the proposed PID algorithm to represent the real commercial PID algorithm, there are three variants to consider for the application of optimal tuning by simulation:

1) To program the PID algorithm in the controller as used in the simulation (if the software license is available).
2) To change the controller (to one that has a known model and repeat the procedure or to one where it is possible to program the PID algorithm and then apply the parameters of the proposed tuning)
3) To apply the tuning anyway on the current controller and evaluate the results.

The mathematical model of the AHU used to validate the closed-loop control system was developed by García et al. in Ref. [34]. The algorithm to simulate this model, as well as the control system, based on the approximation of the transfer functions of the model developed in Ref. [34] using difference equations, is presented in Ref. [35]. The diagram in Fig. 8 shows the interaction of the AHU model with the models of the PID controllers of the internal control loop (see Fig. 3). The model formed by the connection of the models of the AHU and the PID controllers not only allows to validate the control system, but is also the basis of the simulation of this on which the optimal tuning procedure is based.

4.2.2. Application of the multicriteria tuning procedure

According to the procedure proposed in Fig. 2, the parameters to be tuned must first be defined. In this case study, three PID controllers need to be tuned. Since the multivariable process being controlled presents a strong interaction between its variables, the control loops are coupled, which implies that the control actions of a loop to regulate its controlled variable affect the rest of the loops. Therefore, for an optimal tuning of all the controllers, it is proposed that the parameters of the three controllers be tuned simultaneously, to obtain the set of parameters that provides the best performance of the control system.

The controller model is the one shown in Fig. 4, where it can be seen that the parameters to be tuned for each controller are: K_p (proportional gain), T_i (integral time), T_d (derivative time), T_f (time constant of the first order derivative filter), N_z (neutral zone), T_r (rise time) and I_{max} (saturation limit of the integral action). Although the time constant T_t of the anti-windup method used is a tunable parameter, it was preferred to apply the rule of thumb proposed in Ref. [17] to determine its value as a function of the values of T_d and T_i. This simplifies the optimization problem. On the other hand, the I_{max} saturation was added because its effect on the system performance, more pronounced than that of T_t, could be verified during several simulation tests. The ranges and resolutions of the PID controller parameters, selected according to the control system study, are shown in Table 1.

Since the method of aggregating the objectives into a single cost function was selected, the particle swarm optimization (PSO) method can be used. In this case, to limit the solutions to the selected resolution, the method previously used in Ref. [36] was used, which was conceived as an integer PSO method, in which each particle consists of a vector of integers (in this case the parameters of the PIDs). Then, the integer values can be scaled according to the range of values they should have. Fig. 9 gives a detailed description of the PSO method used.

Since seven parameters of each of the three PID controllers shown in Fig. 8 must be tuned simultaneously, the problem size is $N = 21$. A swarm with $M = 150$ particles was used to find the optimal solution. A

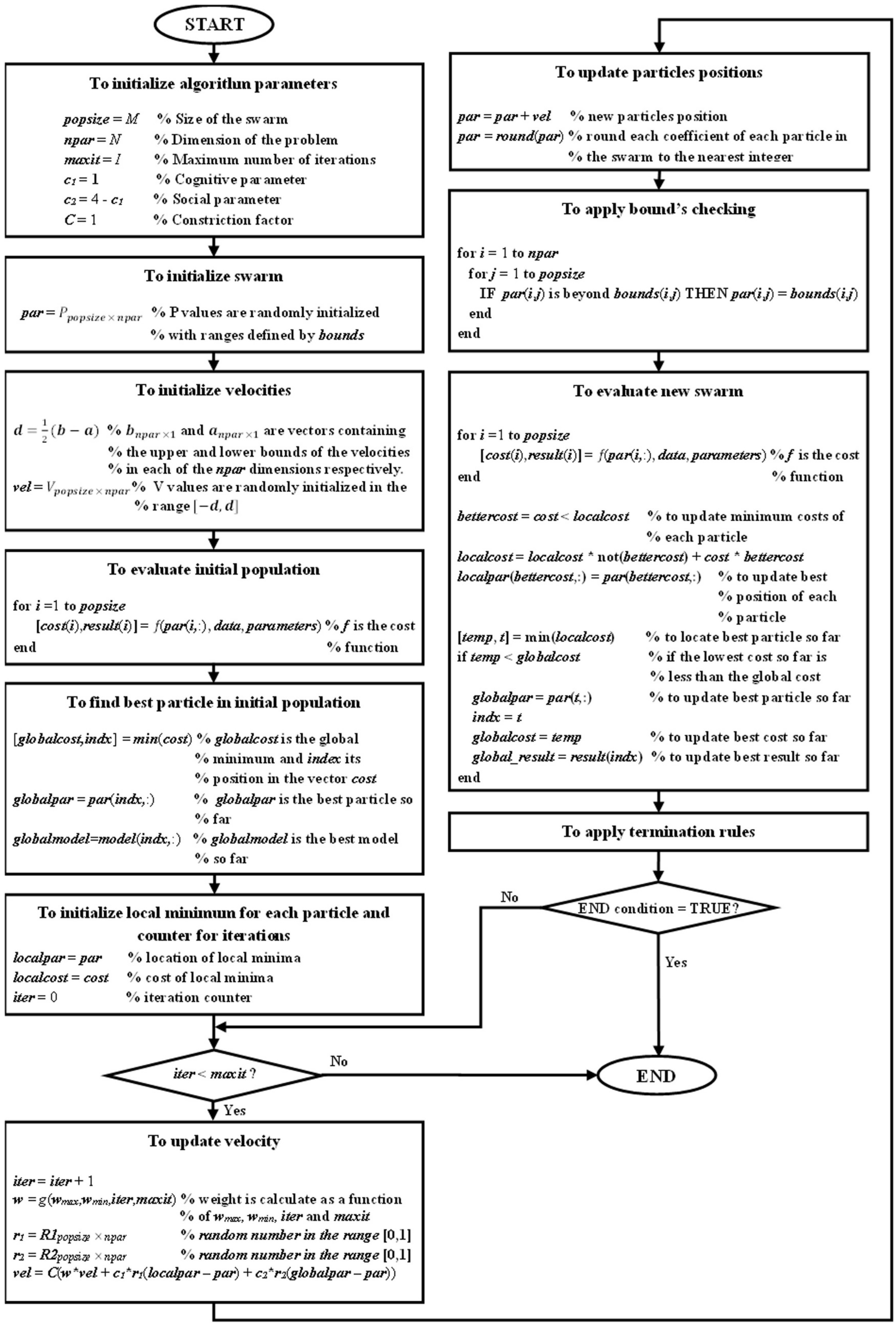

Fig. 9. PSO method used for optimal PID tuning.

Table 2
Relation of cost functions and weights for each experiment.

Experiment	Objectives	J_1	J_2	J_3	J_4	J_5	J_6
1)	**Variables**	ITSEh	ITSEt	–	–	–	–
	Wheights	1	1	–	–	–	–
2)	**Variables**	ITSEh_n	ITSEt_n	–	–	–	–
	Wheights	1	1	–	–	–	–
3)	**Variables**	ITSEh_n	ITSEh_n	–	–	–	–
	Wheights	350	100	–	–	–	–
4)	**Variables**	ITSEh	ITSEt	$ISCO_{heat}$	$ISCO_{cool}$	TV_{heat}	TV_{cool}
	Wheights	1	1	1	1	1	1
5)	**Variables**	ITSEh_n	ITSEt_n	$ISCO_{heat\_n}$	$ISCO_{cool\_n}$	TV_{heat_n}	TV_{cool_n}
	Wheights	1	1	1	1	1	1
6)	**Variables**	ITSEh_n	ITSEt_n	$ISCO_{heat\_n}$	$ISCO_{cool\_n}$	TV_{heat_n}	TV_{cool_n}
	Wheights	350	100	60	50	60,000	400
7)	**Variables**	ITSEh	ITSEt	E_{heat}	E_{cool}	TV_{heat}	TV_{cool}
	Wheights	1	1	1	1	1	1
8)	**Variables**	ITSEh_n	ITSEt_n	E_{heat_n}	E_{cool_n}	TV_{heat_n}	TV_{cool_n}
	Wheights	1	1	1	1	1	1
9)	**Variables**	ITSEh_n	ITSEt_n	E_{heat_n}	E_{cool_n}	TV_{heat_n}	TV_{cool_n}
	Wheights	350	100	60	50	60,000	400
10)	**Variables**	ISTEh_n	ISTEt_n	E_{heat_n}	E_{cool_n}	$TV^2_{heat\_n}$	$TV^2_{cool\_n}$
	Wheights	350	100	60	50	60,000	400

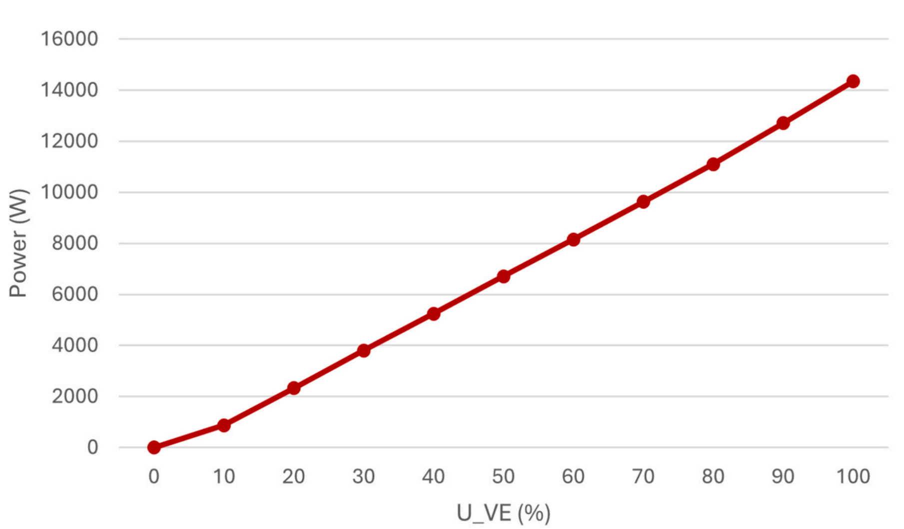

Fig. 10. Curve of measured power values Vs control signal values of the bank of electrical resistors.

maximum number of iterations $I = 100$ was set. The value of w was calculated according to Eq. (8), for which $w_{min} = 0.4$ and $w_{max} = 0.9$ were selected.

$$\omega = \omega_{max} - \frac{iter}{iter_{max}}(\omega_{max} - \omega_{min}) \tag{8}$$

The cost function to evaluate the multiple objectives of interest in the tuning of PID controllers for this case study can be stated as in Eq. (9), where each J_i represents the value of an objective as a function of the set of controller parameters k, and each w_i is the weight associated with that objective J_i.

$$J(k) = \sum_{i=1}^{j} w_i J_i(k); wherek = [K_p, T_i, T_d, T_f, N_z, T_r, I_{max}] \tag{9}$$

Ten cost functions were tested in this work, which are summarized in Table 2, where, for each of the cost functions tested, the objectives considered and the tuned weights for each of them, following the procedure proposed in Fig. 2, are presented. In the first experiment, the performance index ITSE is evaluated for each controlled variable (ITSEh for relative humidity and ITSEt for temperature), since the objectives with the highest priority are the adequate tracking of each controlled variable of its setpoint. In the case of temperature, this index was redefined, considering that this variable must be maintained between two setpoints (SpTSuHi and SpTSuLo). Therefore, while ITSEh is calculated using the expression of Eq. (2), the calculation of ITSEt is done according to Algorithm 1.

Algorithm 1

```
ITSEt = 0;
    eTSuHi = TSu-SpTSuHi              % Compute error with respect to SpTSuHi
    eTSuHi = sat(eTSuHi,0,inf)        % Ensure error is always nonnegative
    eTSuLo=SpTSuLo-TSu                % Compute error with respect to SpTSuLo
    eTSuLo = sat(eTSuLo,0,inf)        % Ensure error is always nonnegative
    for k = 1:L                       % L is the number of simulated values for TSu
      if  SpTSuHi(k) ≥ TSu(k) AND SpTSuLo(k) ≤ TSu(k)        % if TSu is
  between its setpoints
        ITSEt = ITSEt + 0             % ITSEt does not increase
      elseif SpTSuHi(k,2) < TSu(k)    % if TSu is higher than SpTSuHi
        ITSEt = ITSEt+(k)*((eTSuHi^2(k + 1) + eTSuHi^2(k))/2)*Ts)
      elseif SpTSuLo(k) > TSu(k)      % if TSu is lower than SpTSuLo
        ITSEt = ITSEt+(k)*((eTSuLo^2(k + 1) + eTSuLo^2(k))/2)*Ts)
      end
    end
```

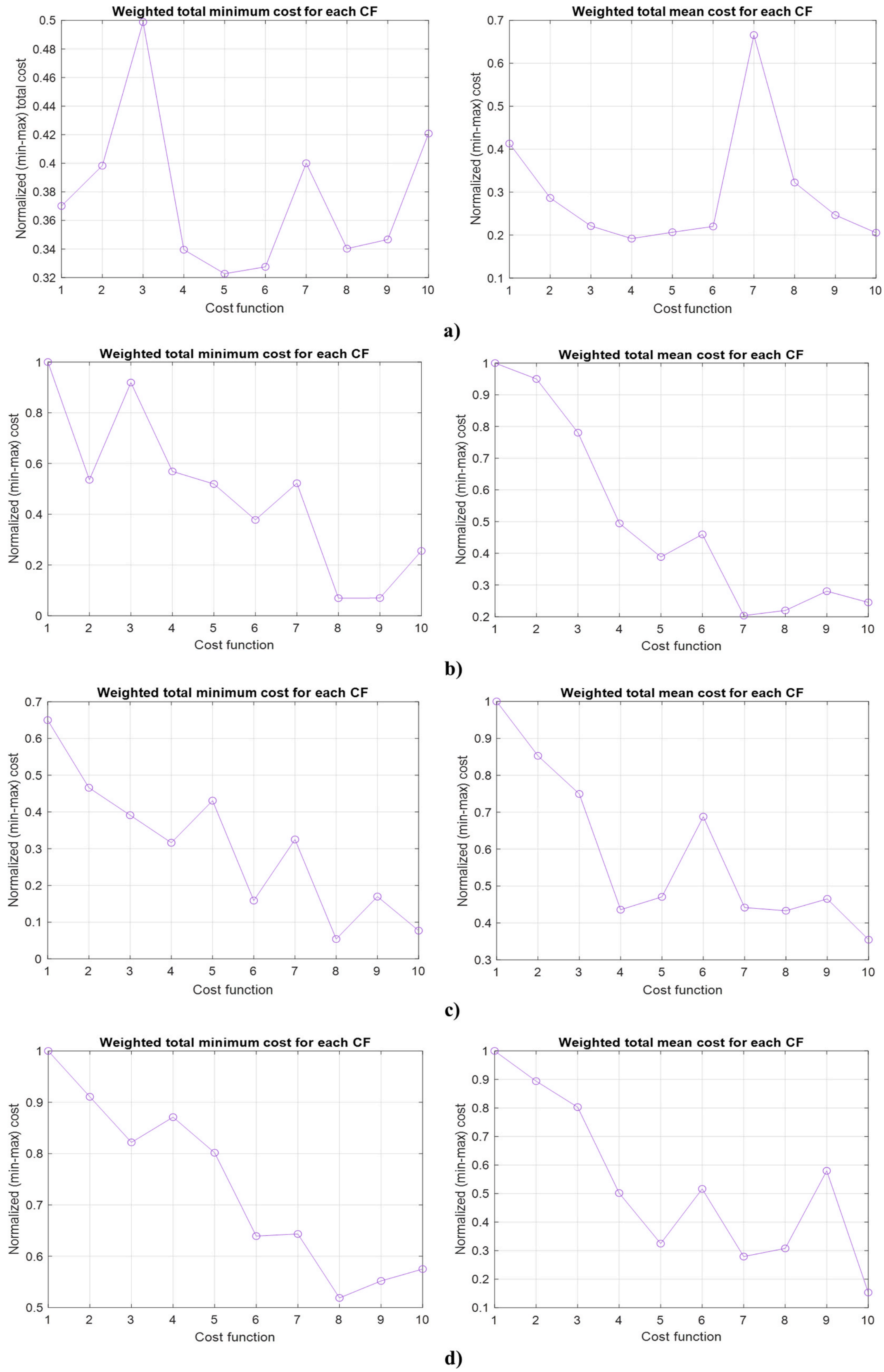

Fig. 11. Evaluation of the weighted mean and minimum cost reached for each cost function after ten tunings for each of the four simulation scenarios: a) scenario 1, b) scenario 2, c) scenario 3 and d) scenario 4.

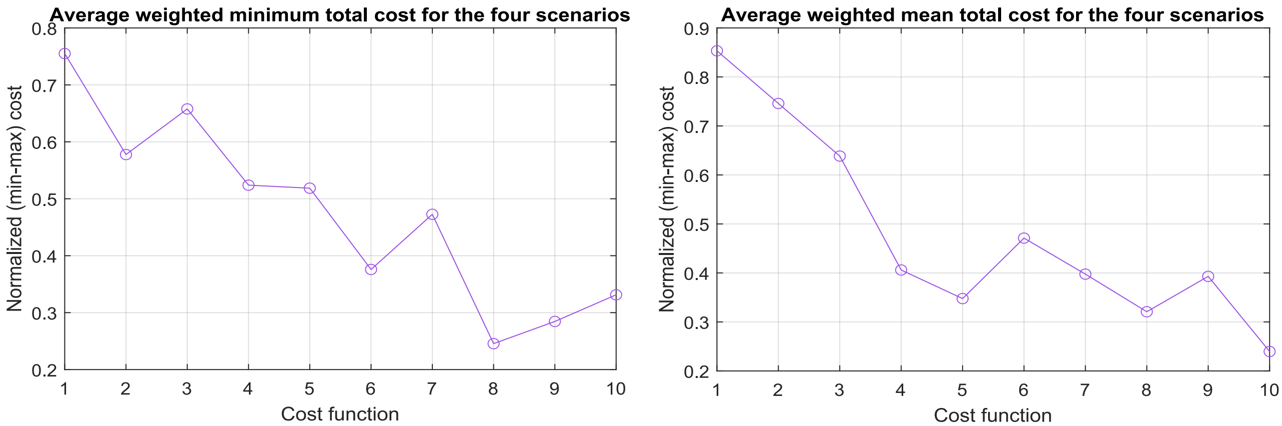

Fig. 12. Average of results for the evaluated scenarios.

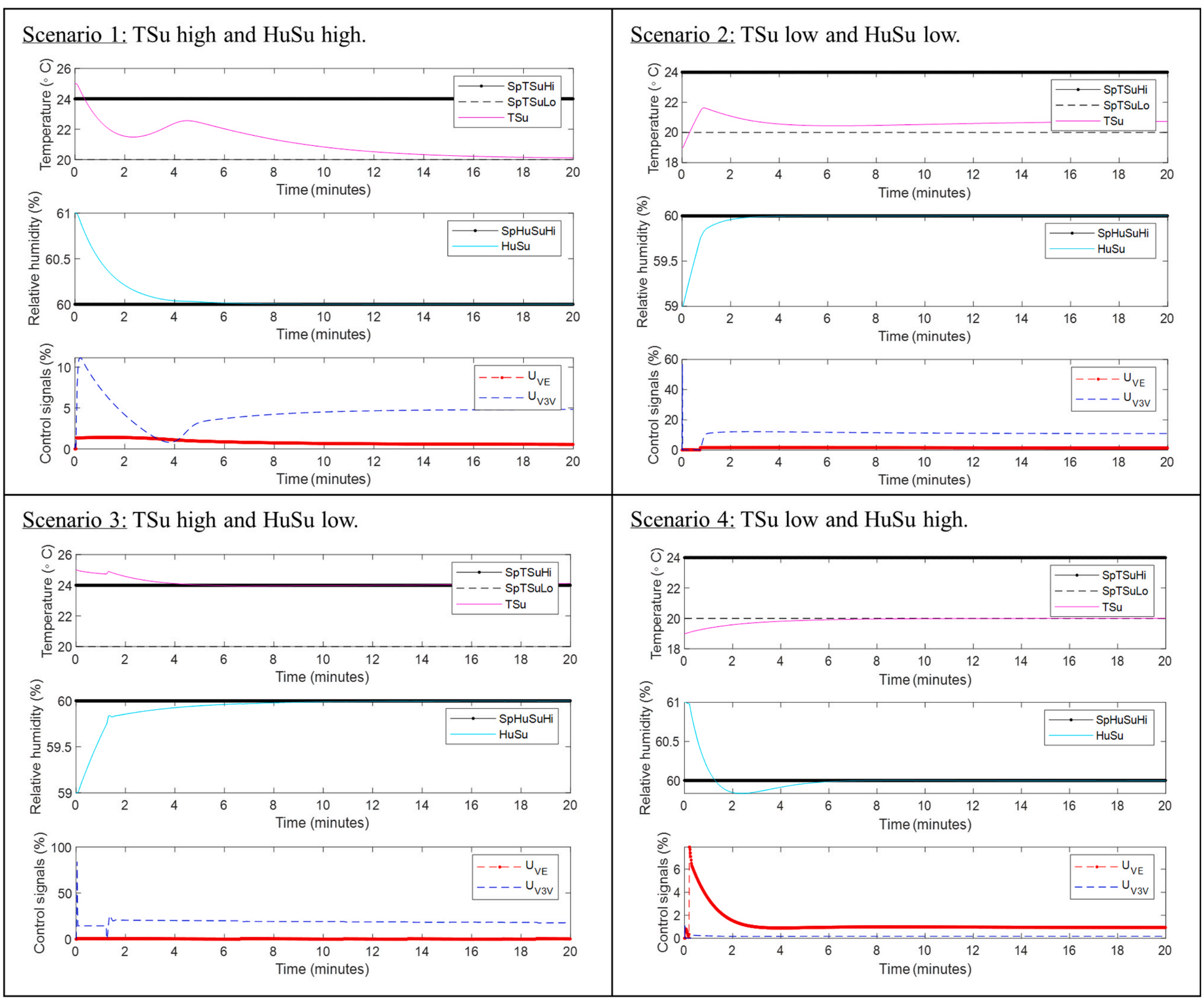

Fig. 13. Best results for each scenario using cost function from tenth experiment.

As explained above, the different ranges of values of the variables, as is the case in this multivariable system, can contribute to giving more weight to some objectives than others, which is why, as a second experiment, it is proposed to scale the values of each objective considered in the first experiment. This paper proposes the following way to scale the objectives: divide the value of each objective by its "worst value". This "worst value" is not necessarily the worst, but a very bad one such that, if the objective J_i has a value lower than $Jworst_i$, it contributes to the cost a value between 0 and 1, if it is equal to $Jworst_i$ the contribution is 1, and if it is higher, the contribution is more than 1.

This scaling also seeks to obtain a cost function whose range of values is not very large, without large changes from one point to another, to

Table 3
Optimal PID parameters for each scenario.

Scenario	Controller	K_p	T_i (min)	T_d (min)	T_f (min)	N_z	T_r (min)	I_{max}
1	Humidity	187	3	0	$1.66667 \bullet 10^{-5}$	2	5	472
	TempLo	877	2	1	$1.66667 \bullet 10^{-5}$	0	5	1000
	TempHi	0.1	4	0	$1.66667 \bullet 10^{-5}$	1	5	50
2	Humidity	196	2	0	6	2	0	50
	TempLo	0.1	4	1	$1.66667 \bullet 10^{-5}$	0	0	50
	TempHi	0.1	9	0	3	1	0	636
3	Humidity	263	7	0	$1.66667 \bullet 10^{-5}$	0	0	70
	TempLo	14	0	1	10	0	0	70
	TempHi	0.10	10	0	10	0	1	761
4	Humidity	0.10	10	0	6	2	3	292
	TempLo	317	10	1	$1.66667 \bullet 10^{-5}$	1	3	971
	TempHi	95	6	0	10	1	1	133

Table 4
Values of the evaluated objectives for each scenario.

Scenario	$ITSE_t$	$ITSE_h$	$E_{cooling}$	$E_{heating}$	$TV_{cooling}$	$TV_{heating}$
1	1.65608	31.37217	397.25168	16.25084	27.83228	2.27841
2	0.89956	7.54885	980.29410	29.33243	127.40829	2.33590
3	80.7442	26.4647	52.4807	0	2.4696	0
4	5.6579	19.6756	0.1080	19.5450	2.2616	1.3860

facilitate the optimization method to find the optimum. A similar strategy of smoothing the objective function was successfully implemented in the application of the PSO method in Ref. [36]. In the case of the ITSE index, here the worst ITSE value is taken as that calculated when the controlled variables remain constant, in other words, the control system fails to modify their values to bring them closer to their setpoints. Of course, there can be infinitely worse cases, in which the error increases. But this case, being unique, allows to calculate a reference value for scaling.

In the third experiment, weights are added to the objectives of the second experiment. In the fourth experiment, objectives associated with the cost of the control are also added. Such as the ISCO index and the TV index. Since the system has two inputs and two outputs, this experiment implies that the cost function adds a total of six objectives. In addition to ISTEh and ITSEt, ISCOheat and TVheat associated with the control signal U_VE, and ISCOcool and TVcool associated wirh the control signal U_V3V are evaluated. No scaling or weights are applied in this experiment. In the fifth experiment, scaling is applied with the same technique. In this case, the worst ISCO is defined as the one that results from a control signal saturated at 100 % during the whole simulation and the worst TV as the one that results from a control signal that at each sampling instant goes from its minimum value to its maximum value and vice versa, depending on where it was at the previous instant, in other words, a totally oscillatory control signal. In the sixth experiment, in addition to the scaling, weights are added to the objectives, being the ITSE indices the highest priority, followed by the ISCO indices and then the TV indices.

The order of priority in tuning the weights is determined by the priorities and practical requirements of the control system. In this case the most important requirement is to ensure the right air temperature and humidity conditions, then to do it with the lowest possible energy consumption and, finally, to avoid as far as possible the wear of the actuators.

The seventh to ninth experiments repeat the same sequence as the fourth to sixth experiments, with the difference that the ISCO index is replaced by an energy consumption model, since this is the variable that is actually to be minimized in the objectives J3 and J4.

The model of energy consumption in the electric heating resistors bank was developed experimentally, following the procedure suggested by Ref. [34]. The results of the experiment measuring the power consumed for a series of values of the control signal U_VE are shown in Fig. 10. A slightly nonlinear behavior can be seen in the U_VE range from 0 to 20 %. The rest of the curve is quite linear, so a model was developed by sections by means of curve fitting, which is shown in Eq. (10). In the case of the cooling coil energy consumption, it is not possible to directly measure the energy consumed at control signal values. In this case, given the information available in the AHU technical documentation, which specifies the maximum power of this coil, a linear relationship between the power and the control signal U_V3V was assumed, giving rise to the model in Eq. (11). The energy consumed during a given period of time is calculated by trapezoidal integration of the power obtained from the models of Eq. (10) and Eq. (11).

$$\begin{cases} P_{heat} = 2.92 \bullet 7U\_VE^2 + 57.73 \bullet U_{VE} - 4.8 \bullet 10^{-14}; & if\ U\_VE \leq 20 \\ P_{heat} = 149.7 \bullet U_{VE} - 774.7; & if\ U\_VE > 20 \end{cases} \tag{10}$$

$$P_{cool} = 270 \bullet U\_V3V \tag{11}$$

For the tenth experiment, a cost function is proposed that instead of using the ITSE index uses the ISTE index [28], which is calculated similarly to the ITSE index but squaring not only the error but also the time. A modified variant of the TV index (squaring it) is also used. This cost function aims to penalize the time it takes for the system to reach the setpoints and stabilize at them, as well as oscillatory control signals. The results of applying the ten cost function variants are presented and discussed in the following section.

5. Results and discussion

To evaluate the cost functions proposed in the previous section, the PID controllers were tuned for the followin scenarios:

1) TSu and HuSu higher than the values of SpTSuHi and SpHuSuHi respectively.
2) TSu and HuSu lower than the values of SpTSuLo and SpHuSuHi respectively.
3) TSu higher than SpTSuHi and HuSu lower than SpHuSuHi.
4) TSu lower than SpTSuHi and HuSu higher than SpHuSuHi.

For each scenario, the controllers were tuned ten times with each cost function, in order to compare the results and analyze some statistics that allow conclusions to be made. For the evaluation of the results, a

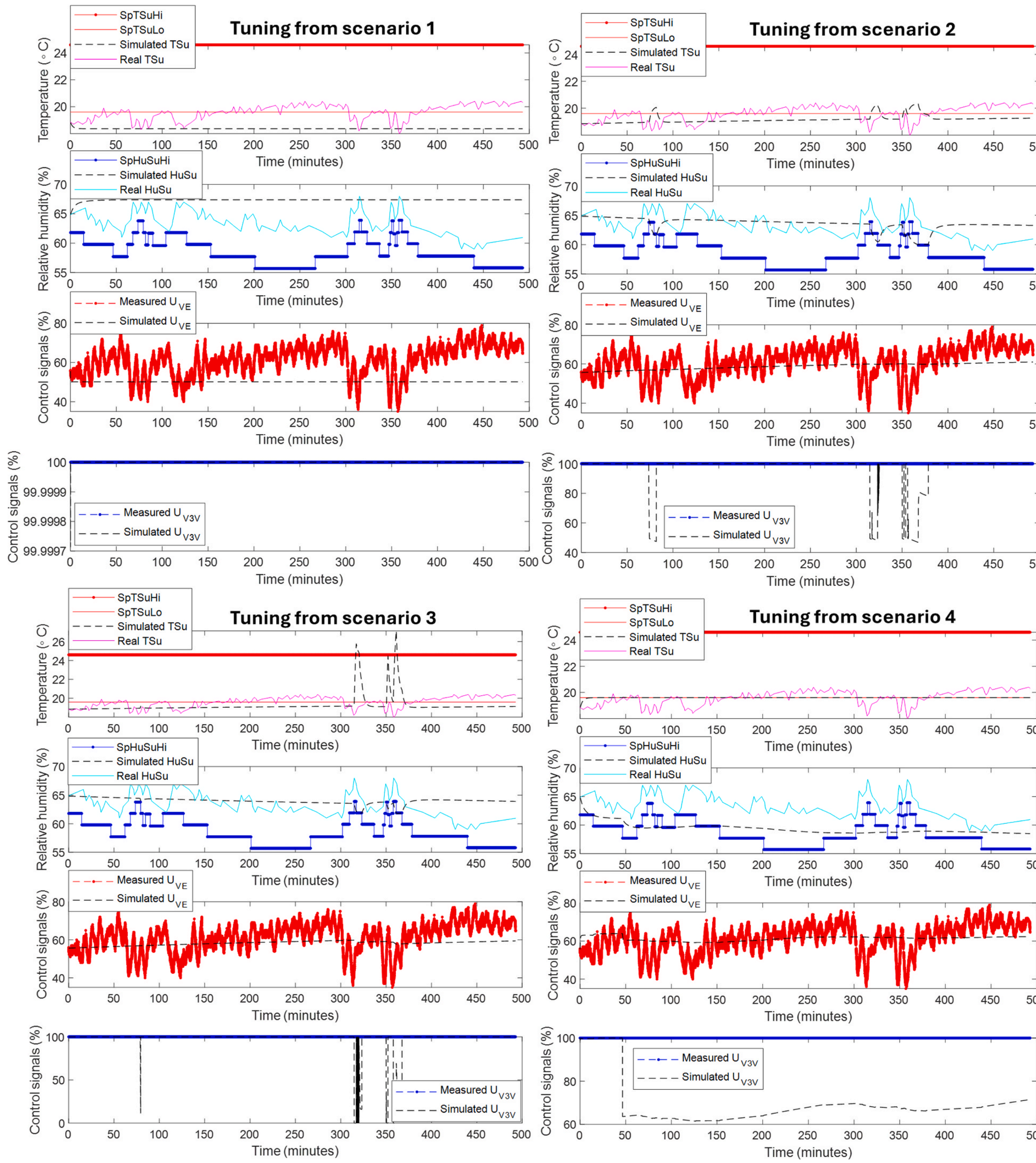

Fig. 14. Application of the best tuning calculated in each of the above scenarios to a real control scenario.

Table 5
Results of optimal tuning for a real control scenario.

Case	ITSEt	ITSEh	Energy for heating (kW/h)	Energy for cooling (kW/h)	Total energy (kW/h)	$TV_{heating}$	$TV_{cooling}$
Real	$1.04551 \bullet 10^6$	$1.82103 \bullet 10^8$	68.91548	221.75847	290.67395	9811.50374	0
Simulation	84.65501	$4.32439 \bullet 10^7$	69.09761	153.68554	222.78315	22. 76,509	65.704
Difference	$1.04543 \bullet 10^6$	$1.38859 \bullet 10^8$	−0.18213	68.07293	67.89080	9788.73865	−65.704
Percentage	99.99 %	76.25 %	−0.26 %	30.69 %	23.35 %	99.76 %	—

cost function was constructed with the same objectives evaluated in the seventh through ninth experiments (ITSEh, ITSEt, Eheat, Ecool, TVheat and TVcool). In this case, different weights were chosen for assessing the evaluation of the results. While the weights in the optimization had the function of compensating and helping obtain results that favored the highest priority objectives, they also served to shape the cost function in order to make it easier for the optimization method to arrive at good solutions. At the stage of evaluation of the results, the weights aim to assess a value to the fulfillment of each objective that allows to reach a kind of integral qualification of the result. Thus, for the evaluation of the results, a set of weights was chosen whose sum is one, so that each weight assigned to an objective confers a part of the total value of the cost. The objectives were also scaled to the range of 0 to 1, applying Eq. (12) [37], which facilitates their comparison and aggregation to perform the integral evaluation of the result. Fig. 11 shows the results of the evaluation with this criterion of the different cost functions for each of the 4 simulation scenarios tested.

$$J_{nm} = \frac{J - J_{min}}{J_{max} - J_{min}} \tag{12}$$

When analyzing the scenarios separately, it is not clear which cost function variant offered the best overall results. However, it can be seen in Fig. 12 that the mean of the average costs of the four scenarios was lower using the cost function proposed in the tenth experiment. In general, it can be observed that the use of the energy measures as part of the cost functions allowed better results to be obtained with respect to the cost functions using the ISCO index and better, in both groups of cost functions, than the results obtained using functions based only on time response indices.

Fig. 13 shows the results of simulating the control with the best tunings achieved by using the tenth cost function evaluated for each of the four simulation scenarios. The sets of tuning parameters for each controller in each scenario are summarized in Table 3. It can be seen that the time response of the system in all four scenarios is good: fast, with no overshoot or oscillations. This was achieved with low power consumption, as shown in Table 4 and mostly smooth control signals.

5.1. Evaluation of tuning in a real simulation scenario

Fig. 14 shows the results of applying the four tunings obtained in the four simulation scenarios previously evaluated to a real simulation scenario, with data obtained from the process using the SCADA system. As can be seen, this real scenario is very similar to the fourth simulation scenario previously evaluated (TSu starts lower than SpTSuLo and HuSu starts higher than SpHuSuHi). The tuning of the fourth scenario yielded a better result for the actual simulation scenario than that offered by the control system currently in operation. Table 5 summarizes the results of applying the tuning of the fourth simulation scenario to the real scenario. It highlights that the ITSE of the temperature improved by more than 99 % and the ITSE of the relative humidity by more than 76 %. The heating coil consumed 0.26 % more energy, but the cooling coil consumed more than 30 % less, so the overall AHU energy consumption was reduced by 23.5 %, with much better reference tracking and more reliable control signals overall. Furthermore, during the tuning experiments, the execution time of the optimization algorithm was recorded, and it was found that obtaining the 21 optimal parameters of the three controllers simultaneously took between one and four minutes.

However, it is a limitation that depending on the operating point of the system, when changing, it may require a retuning of the controllers. Future work on this case study should research whether it is possible to obtain a tuning that achieves a compromise in system performance across different operating points or whether it is necessary to try alternatives such as gain scheduling.

Regardless of the future challenges that remain in the optimization of this case study, starting with the practical application of the obtained tuning, the general method for dealing with complex commercial controllers proved, at least at the simulation level, its validity, as well as the rest of the proposed procedures, including the proposed cost function, which outperformed the other nine with which it was compared, which include objectives frequently evaluated in literature. Thus, pending practical validation, the simulation results point to the advantages of including energy consumption models in the cost function. For future work, also the multiobjective Pareto approach must be evaluated in this system.

6. Conclusions

The application of the general method proposed for the tuning of commercial PID controllers, proved its validity by being able to meet the aim of the case study presented: to improve the current control of temperature and humidity with lower energy consumption. Although an efficiency metric has not been calculated, it can be affirmed that, as the controller with the calculated tuning achieves a better tracking of the setpoints and invests less energy for it, it is consequently more energy efficient. It was proved that by applying the proposed procedures, the task of tuning complex controllers with a large number of parameters simultaneously in multivariable control systems with coupled control loops could be carried out with a relatively low computational cost (taking into account the algorithm execution time). The main limitation of this work is that the results have only been validated through simulation. If put into practice, the greatest chances of success are in the programming of the control algorithm used for the simulations, since the validation of the PID controller model for the PLC, with unknown PID algorithm, was not satisfactory. As for the general tuning method proposed, it has the limitation of needing models of both the process and the controller, which are not always easy to obtain. As a suggestion for future work, the Pareto approach should be applied and results compared, as well as other objectives, since the comparisons made in this work are still limited.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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